

Group Members: David Hong, Farhen Shefa, Yanka Sikder

CS-GY 6053 Foundations of Data Science (F24)

Professor Reeves

Final Project Proposal

1. Identify a Question/Estimand

Question: Does higher Gross Domestic Product (GDP) per capita causally affect higher life expectancy?

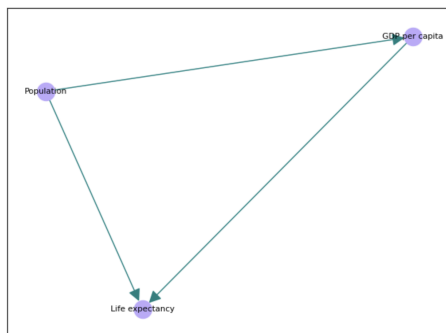
2. Identification and Description of the data to be used:

The dataset we are using is the Gapminder dataset (<https://www.gapminder.org/data/>) by Jennifer Bryan which contains data on life expectancy and GDP per capita by country and year. It contains 1704 observations from 2022 and contains 6 variables: *country*, *continent*, *year*, *life expectancy*, *population*, and *gdpPercap*. Since we are only allowed to use three variables, we will be using *pop* - population, *gdpPercap* - GDP per capita (US\$, inflation-adjusted), *lifeExp* - Life expectancy at birth, in years. We will be utilizing the data from this csv file:

<https://github.com/NickCH-K/causaldata/blob/main/Python/causaldata/gapminder/gapminder.csv>

3. Casual Model in the form of a directed acyclic graph (DAG):

The casual model has three variables including a **treatment variable**: *gdpPercap* - GDP per capita (US\$, inflation-adjusted), **outcome variable**: *lifeExp* - Life expectancy at birth, in years, and **potential confound variable**: *pop* - Population. We are assuming that GDP per capita can influence life expectancy, as wealthier countries typically have better healthcare, and better access to nutritious food and clean water, so better life expectancy. However, population size can affect both GDP per capita and life expectancy. Larger population can influence the economic productivity of individuals, as more people contribute to the workforce, which can increase the GDP per capita. Population also can influence life expectancy – as population increases, competition for resources increases, which decreases resources like healthcare, food, etc. available per individual, which can then decrease life expectancy.



4. Define a Corresponding Statistical Model

Bayesian Linear Regression might be the suitable model for analyzing the relationship between life expectancy, GDP per capita, and population. This model allows us to incorporate priors about expected information such as higher GDP per capita typically correlates with higher life expectancy. Adjusting confound variables is another beneficial aspect of Bayesian Linear Regression model.